Thiago Ricardo

Santa Cruz, CA | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Embedded systems engineer (M.Sc. Cybersecurity Engineering candidate, Politecnico di Torino) with hands-on projects in multi-process C/POSIX systems, CAN intrusion detection, and CI-tested embedded software. Postgraduate training in automotive software (UFPE): ECU architecture, CAN/LIN/FlexRay, Automotive Ethernet, AUTOSAR, ISO 26262, and RTOS/QNX. Authorized to work in the U.S.; open to remote and Bay Area roles.

SKILLS

Languages: C, C++, Python, SQL
Embedded & OS: POSIX (processes, IPC, pthreads), RTOS scheduling, QNX, OSEK-VDX (Trampoline), Arduino, GPIO/interrupts, ECU architecture
Automotive: CAN, DBC, LIN, FlexRay, Automotive Ethernet, AUTOSAR (basics), ISO 26262 (overview), Automotive SPICE, V-model
Security: CAN Intrusion Detection System (IDS), MITRE ATT&CK, penetration testing (coursework), vehicular cybersecurity fundamentals
Tools & DevOps: Linux, Git, Docker, GitHub Actions (CI/CD), Make, LCOV, Cppcheck, pytest, python-can, cantools
Data & ML: scikit-learn, pandas, NLP (TF-IDF), Matplotlib

PROJECTS

Vehicle Management Unit (VMU) for Hybrid Vehicles https://github.com/thiagoricardop/VMU-for-Hybrid-Vehicles 2025
C, POSIX, Docker, Make, LCOV, Cppcheck, GitHub Actions · STELLANTIS Technical Residence (UFPE), team of 4

- Developed a multi-process C simulation of a hybrid powertrain in which a VMU coordinates electric and combustion engine modules through commands and shared data, with graceful shutdown on SIGINT/SIGTERM.
- Automated build, unit tests, and LCOV coverage reports with a Makefile inside a Docker environment.
- Integrated Cppcheck static analysis and test/coverage workflows into GitHub Actions CI.

CAN Bus Sniffer, DBC Parser & Intrusion-Detection Demo https://github.com/thiagoricardop/can-bus-sniffer-parser Oct 2026
Python, python-can, cantools, pytest, GitHub Actions

- Built a CAN sniffer that decodes raw frames into named messages and physical signal values using DBC files on virtual and SocketCAN interfaces.
- Implemented a rule-based IDS detecting unknown arbitration IDs (spoofing), out-of-range signal values (tampering), and frame flooding (denial of service).
- Developed a traffic simulator with attack injection, CSV/JSON Lines logging, pytest tests, and GitHub Actions CI with coverage on Python 3.10-3.12.

SSH Honeypot Attack Classification (ML & MITRE ATT&CK) github
Python, scikit-learn, pandas · ML4N course, team project · Politecnico di Torino

- Analyzed approximately 230,000 SSH honeypot shell sessions and mapped them to seven MITRE ATT&CK-based intent classes (multi-label).
- Built Bag-of-Words/TF-IDF feature pipelines and compared supervised classifiers and clustering approaches in reproducible notebooks.

EXPERIENCE

Systems Engineer Intern Nov 2023 – Apr 2024
SYSTRA Brazil *Belo Horizonte, Brazil*

- Structured and governed the document database for the Guadalajara light-rail project, improving traceability of engineering deliverables across stakeholders.
- Built BI dashboards in Python and Power BI that exposed design-phase bottlenecks for root-cause analysis.
- Accelerated the Design Stage by 50%, completing it ahead of schedule.

Junior Software Developer (part-time) Apr 2023 – Jul 2023
IT4Innovations National Supercomputing Center *Ostrava, Czechia*

- Developed and maintained a Python GUI to visualize job and resource data on the Karolina and Barbora supercomputers in a Linux/HPC environment.
- Redesigned the UX/UI based on user feedback; improvement confirmed by 95%+ of employees who used the tool.

Photovoltaic Project Intern Oct 2021 – Mar 2023
Ciber Solar *Pará de Minas, Brazil*

- Produced 3D layouts, electrical schematics, and Bills of Materials for 200+ photovoltaic plant projects with a 95%+ approval rate.

EDUCATION

Politecnico di Torino Italy
M.Sc., Cybersecurity Engineering. *Sep 2027 (expected)*

Relevant Coursework: Machine Learning for Networking, Penetration Testing

UFPE Brazil

Postgraduate Residency, Automotive Software Development. *Aug 2025*

Relevant Coursework: Embedded Systems, Automotive Networks (CAN/LIN/FlexRay/Ethernet), RTOS and OSEK-VDX, POSIX and QNX, AUTOSAR, Automotive SPICE and ISO 26262, Software Testing and Test Automation, Requirements Engineering

UFMG Brazil

B.Sc., Electrical Engineering (Computer Science emphasis). *Sep 2024*

VSB - Technical University of Ostrava Czechia

Exchange, Computer Science. *Aug 2023*

CERTIFICATIONS

- IELTS Academic (Jan 2025)
- Python Essential Training (2022)
- SQL Essential Training (2019)

LANGUAGES

- English (full professional, IELTS 7.0), Portuguese (native), Italian (intermediate), Spanish (intermediate)

VOLUNTEERING

- Summer Camp Coordinator (Sports), YMCA, Camp Whittle, Big Bear Lake, CA: May 2026 - Sep 2026; led a team and expanded the activity program.
- Camp Counselor, YMCA, Camp Whittle, Big Bear Lake, CA: May 2025 - Sep 2025.